

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2 HT	(a)	(i)		Trachea (1) Intercostal muscles (1) Diaphragm (1)	3					
		(ii)		{Intercostal muscles /B} {contract/pull up}/expand the ribcage (1) {Diaphragm/C} {contracts/lowers/flattens} /increases volume of chest (1) Air pressure inside chest lowers (1)	3					
	(b)			Breathing rate drops for both (1) At 8 minutes difference is 3.6/4 breaths/minute OR At 12 minutes difference is 4.4/5 breaths/minute (1) Matthew's breathing rate decreases quicker / Matthew's breathing rate returns to normal sooner (1)		3			1	3
	(c)			Blood arriving in the alveoli has a higher carbon dioxide concentration than the air in the alveoli (1) There is a (concentration gradient which allows) carbon dioxide to <u>diffuse</u> out of the blood and into the alveolar air. (1*) Similarly, blood arriving in the alveoli has a {lower oxygen / deoxygenated} concentration than the air in the alveoli. (1) Therefore, oxygen moves into the blood by <u>diffusion</u> . (or 1*) <i>Either of the (1*) statements</i>	3					
				Question 2 total	9	3		12	1	3